

Welcome to the second



2025
OPTIMIST

workshop on **open tools**,
interfaces, and **metrics**
for **implementation**
security testing



Speakers!
Please upload
your slides

Welcome to OPTIMIST

September 14-18, 2025
Kuala Lumpur, Malaysia

Organizers

- Aydin Aysu, NCSU
- Fatemeh Ganji, WPI
- Patrick Schaumont, WPI
- Caner Tol, WPI

TPC

- Lejla Batina (Radboud University)
- Gaëtan Cassiers (UCLouvain)
- Daniel Dinu (Intel)
- Kris Gaj (George Mason University)
- Daniel Page (University of Bristol)
- Markku-Juhani O. Saarinen (Tampere University)
- Mirjana Stojilovic (EPFL)
- Marc Witteman (Keysight Technologies)

This workshop is for you



September 14-18, 2025
Kuala Lumpur, Malaysia

- **for the learner**

- who is stuck in reproducing experimental results ...
- who wonders how to explain a cool result to the community ...
- who needs a standard dataset, metric, reference implementation ...

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- **for the doer**

- who needs an apples-to-apples comparison in security testing ...
- who needs a new standard dataset, metric in support of a secure product ...

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- **for the learner**

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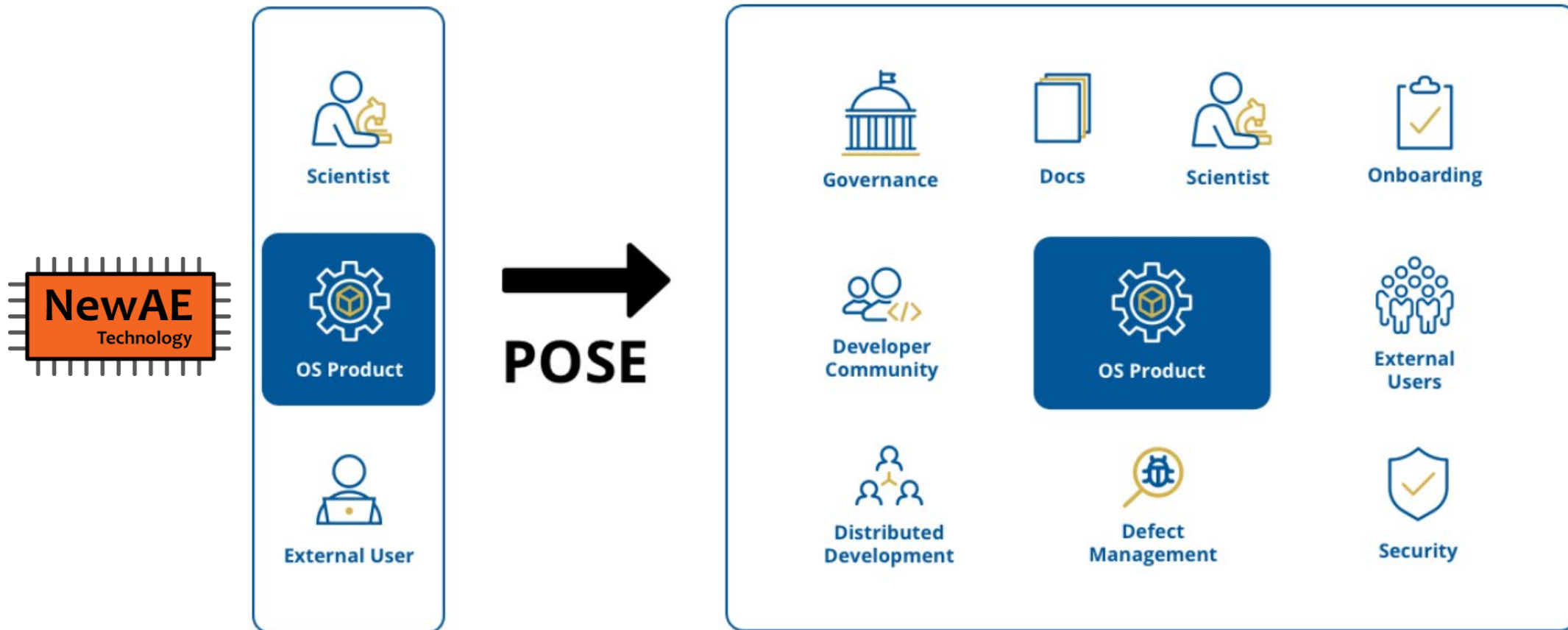
- **for the doer**

- who needs an apples-to-apples comparison in security testing ...
- who needs a new standard dataset, metric in support of a secure product ...

- **for everyone**

- **who believes that better reproducibility in security testing also means better research and better products for everyone**

- 6/24: NSF Pathways to Open Source Ecosystems



Optimist Working Groups

September 14-18, 2025
Kuala Lumpur, Malaysia

File Format For Traces



With contributions from
Aydin Aysu (NC State University)
Gaetan Cassiers (Université Catholique de Louvain)
Fateme Garji (Worcester Polytechnic Institute)
Vincent Immler (Oregon State University)
Jens Peter Kaps (George Mason University)
Trey Marcantonio (Worcester Polytechnic Institute)
Jean-Michel Picod (Google)
Patrick Schaumont (Worcester Polytechnic Institute)
Aurelien Vasselie (eShard)

WORKING DOCUMENT
FILE FORMAT FOR TRACES: REQUIREMENTS AND GLOSSARY

This document is a list of common terms and requirements relevant to a file format for measurement data from implementation security testing, with an emphasis on data that represents side-channel leakage

Version History

0.5	1/6/25	First version based on Optimist Hour Meeting 4/10/24
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Capture Interface for SC & FI Setup



With contributions from
Aydin Aysu (North Carolina State University)
Gaetan Cassiers (Université catholique de Louvain)
Daniel Dinu (Intel)
Anuj Dubey (AWS Amazon)
Fateme Garji (Worcester Polytechnic Institute)
Benoit Gérard (Agence nationale de la sécurité des systèmes d'information--ANSSI)
Thomas Hiscock (Laboratoire d'électronique des technologies de l'information--CEA-Leti)
Jens-Peter Kaps (George Mason University)
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Mirjana Stojković (École Polytechnique Fédérale de Lausanne--EPFL)
Emanuele Strieder (Fraunhofer Institute for Applied and Integrated Security AISEC)
Jean-Pierre Thibault (NewAE Technology Inc.)
Caner Tol (Worcester Polytechnic Institute)
Marc Witterman (Keysight Technologies, Inc.)

WORKING DOCUMENT
CAPTURE INTERFACE FOR SIDE-CHANNEL AND FAULT INJECTION SETUPS

This document provides information about existing capture interfaces to suggest evaluation criteria and recommended features useful for implementation security testing.

Version History

0.5	3/6/25	First version based on Optimist Hour Bi-weekly Meetings from 1/23/25 to 3/6/2025
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Testing PQC Impl Security



Aydin Aysu (North Carolina State University)
Daniel Dinu (Intel)
Kris Gaj (George Mason University)
Fateme Garji (Worcester Polytechnic Institute)
Mona Hashemi (National University of Singapore)
Renita J (Society for Electronic Transactions and Security, India)
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WORKING DOCUMENT
TESTING POST-QUANTUM CRYPTOGRAPHY IMPLEMENTATION SECURITY

This document consolidates community-driven insights and recommendations for implementation security testing of post-quantum cryptography, detailing essential terminology, open-source tools, hardware/software implementations, and strategies for creating reproducible side-channel datasets tailored to the unique challenges of PQC algorithms.

Version History

0.5	6/4/25	First version based on Optimist Working Group Meetings
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Acceleration of AI for Impl Sec Testing



Aydin Aysu (North Carolina State University)
Lejla Batina (Radboud University)
Eswari Devi N (Society for Electronic Transactions and Security - SETS, India)
Daniel Dinu (Intel)
Fateme Garji (WPI)
DebdEEP Mukhopadhyay (IIT KGP)
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WORKING DOCUMENT
ACCELERATION OF AI FOR IMPLEMENTATION SECURITY TESTING

This document reviews the state of open tools, data and methods related to the use of AI acceleration platforms and AI algorithms for implementation security testing. The document also identifies the key areas for improvements and potential for standardization. Previous OPTIMIST documents already discuss the file format for side-channel traces, the capture interface, and PQC testing methods for implementation security testing campaigns.

Version History

0.5	7/8/25	First version based on Optimist Working Group Meetings
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- 4 OPTIMIST Hours per working group/document
- 30 Unique Document Contributors

Optimist Working Groups

September 14-18, 2025
Kuala Lumpur, Malaysia

File Format For Traces



Open Tools, Interfaces and Metrics for Implementation Security Testing
<https://optimist-ose.org/>

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<https://www.optimist-ose.org>

Program - <https://optimist-ose.org/ches25>

September 14-18, 2025

Time	Topic
9:15	Mukhopadhyay on AI
9:30	Sullivan on SCARR
9:45	Invited: Gierlichs on ORSHIN
10:30	Coffee Break
11:00	Spotlight on Artifacts I
11:30	Krieger on OpenNTT
11:45	Eisenbarth on MicroWalk
12:00	Zhou on Verification
12:15	Renita on ISO 17825
12:30	Lunch

OPTIMIST 2025 statistics

- 11 talk submissions, 7 accepted
- 9 artifact spotlight submissions, 9 accepted

Time	Topic
14:00	Invited: Kannwischer on PQC
14:45	Ganji on SCAPEGoat
15:00	Spotlight on Artifacts II
15:30	Coffee Break
15:00	Panel on Governance Hutter, Riviere, Schwabe, Witteman



Speakers!
Please upload
your slides

Enjoy the Workshop!

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- **Participate in OPTIMIST**

- Engage with Speakers
- Participate in Working Groups
- Propose New Working Group Topics
- Contribute New Artifacts, Datasets, Metrics



- **Become an Optimist**

- Join Now on <https://optimist-ose.org>

- **Thank you, TPC, Speakers and Panelists!**